Lightweight and Efficient Multimodal Prompt Injection Network for Scene Parsing of Remote Sensing Scene Images

Yangzhen Li, Wujie Zhou, *Senior Member, IEEE*, Jiajun Meng, and Weiqing Yan, *Senior Member, IEEE*

***Abstract*— Scene parsing of high-resolution remote sensing images with complex backgrounds has received extensive attention in recent years. As unimodal networks are significantly affected by weather conditions, reflecting complex ground conditions fully and accurately is difficult; therefore, multimodal scene analysis is particularly important. Current multimodal scene-parsing networks often employ a dual-coding architecture to achieve high-performance segmentation. Because prompt learning allows models to understand and capture contextual information more effectively, the proposed prompt injection module (PIM) extracts relevant information from frozen normalized digital surface model (nDSM) features and integrates it into the infrared, red, and green (IRRG) branches through a modal embedding block. To extract the contextual semantic relationships between the local and global features in the image efficiently, we also design a dynamic filter block for feature enhancement. This design facilitates the mutual complementarity and guidance of information between the two modalities and optimizes fusion. The experimental results demonstrate that lightweight and effective multimodal prompt injection network (LENet) outperforms most current state-of-the-art lightweight methods on two public datasets, achieving comparable accuracy to that of traditional methods. It has only 10.81 M parameters, with 2.72 GFLOPS. Our code and results are available at https://github.com/LYZ00918/LENet.**

***Index Terms*— Dynamic filter block, multimodal, prompt injection network, remote sensing.**

I. Introduction

SCENE-PARSING interprets each pixel value in an input image and classifies it. Recent developments in deep learning and advancements in unmanned aerial vehicle technology have expanded the scenarios for applying scene parsing to remote sensing images, particularly in land management and planning [1], intelligent transport [2], and agricultural monitoring [3].

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To date, researchers have primarily focused on understanding the semantic scenes of simple images derived from unimodal RGB sources and have achieved relatively satisfactory results. Full convolutional neural networks (CNNs) [4] were the first to be used for scene-parsing tasks and exhibited good segmentation performance. Subsequent refinements have led to the proposal of U-shaped architecture [5] and coding and decoding [6] architectures. Zhao et al. [7] devised a pyramid parsing architecture to capture multiscale contextual information. Other researchers have used global or local attention mechanisms to capture the most interesting parts of a feature graph. For example, Wang et al. [8] used attention mechanisms to capture local features. Wu et al. [9] introduced a self-attention approach to capture global features. Others [10], [11], [12] have also implemented operations such as graph convolution to integrate global contextual semantic relations. Li et al. [13] used boundary information to improve the semantic segmentation of remote sensing images. However, when parsing remote sensing images with complex backgrounds, multimodal networks can provide additional information that can help to improve the segmentation accuracy.

Current multimodal scene-parsing approaches have received considerable attention for exploiting the complementary information between different modal features. Earlier multimodal fusion methods used simple addition, subtraction, and multiplication for modal interaction and outperformed unimodal networks. However, these methods do not fully utilize the potential of the additional modality. Thus, Zhou et al. [14] designed a cross-level multiscale similarity refinement module for the continuous cross-interaction of bimodal information to provide feature complementarity. Similarly, Jin et al. [15] introduced boundary information for locating salient spaces during modal fusion to optimize the fusion accuracy. Several multimodal fusion architectures have been proposed to improve scene parsing for images. A significant drawback of most high-performance segmentation networks is their reliance on numerous parameters and extensive computations, which makes them difficult to deploy on lightweight embedded devices. Zhong et al. [16] and Lv et al. [17] proposed lightweight networks to weigh the relationship between efficiency and lightness, but their performance remains extremely limited. Therefore, we propose a lightweight and effective multimodal prompt injection network (LENet). A unique prompt injection module (PIM) injects frozen normalized digital surface model (nDSM) modalities into the infrared,

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red, and green (IRRG) information stream. Meanwhile, model embedding blocks (MEBs) with few parameters are introduced to realize two mutual complementarities and guidance. This architecture accelerates the training speed and improves the segmentation.

The second challenge facing scene parsing is the contextual complexity of remotely sensed images, which, at small scales, exhibit continuous and varying patterns in categories such as vegetation and cars. Therefore, multiscale contextual information must be captured effectively. Liu et al. [18] utilized deep depth-supervised hierarchical branching to capture information at a specific granularity for each branch. Gao et al. [19] designed an adaptive spatially labeled Transformer encoder to balance its ability to learn remote relations. Feng et al. [20] used channel relations to obtain global multimodal fusion relations for obtaining more detailed contextual semantic information. Zhou et al. [21] used progressive complementary fusion blocks to augment deep features with high-level semantic information for training shallow details. Similarly, Liang et al. [22] used an attention mechanism to extract complementary fusion modal features adaptively.

In this context, we designed a dynamic filter block that first performs a channel shuffle operation on the features to filter them from the channel perspective. Subsequently, the features are sliced according to high and low frequencies to obtain four feature blocks. Multiple convolutional branches with different receptive fields are then used to increase the receptive scale. Unlike in the conventional scheme, we modify the convolution structure by abandoning the 3×3 convolution kernel and using a two-stage 1×3 and 3×1 approach, with two convolutional layers in series. This procedure extracts features in the horizontal and vertical directions. The semantic features that are distributed in multiple directions are aggregated as comprehensively as possible. Finally, richer and more effective representations can be obtained.

The main contributions of this study are summarized as follows.

1) The proposed LENet is lightweight and efficient. More accurate and richer semantic information can be injected from the frozen nDSM modality into the IRRG branch through the PIM architecture. Then, the MEB is used to realize the complementarity of the bimodal information. The entire network has only 10.78 M parameters and 2.1 GFLOPS, with performance close to that of existing methods.
2) We designed an injection block for the multimodal injection method. The nDSM feature information of the frozen middle three layers is fused with the trainable IRRG branch. The discrete cosine transform (DCT) is used to extract the high- and low-frequency information of the nDSM modes to assist with IRRG feature localization. The enhanced fusion features are then re-injected back into the IRRG backbone.
3) To address inconsistent class information in complex backgrounds, we also designed a dynamic filter block, which uses two-stage directional expansion convolution to mine the contextual semantic information of the horizontal and vertical axis directions to improve the receptive field range of the entire network.

The remainder of this article is organized as follows. Section II provides an overview of related studies on multimodal semantic segmentation and prompt injection networks. Section III describes the proposed methodology, including the designed modules and prompt injection framework. Section IV details the experimental approach and results, and Section V summarizes the study.

II. Related Work

A. Multimodal Remote Sensing Image Semantics

It has recently become possible to process remote sensing images with simple backgrounds using CNNs. For example, Li et al. [23] designed a single-modal network for remote sensing images. Lu et al. [24] spliced CNN and Transformer backbones to extract complementary knowledge to improve the segmentation effect. With the multisourcing of modal information, the use of multimodality to understand remotely sensed scenes has received increasing attention. For example, Kampffmeyer et al. [25] simply summed multimodal feature information to enhance the complementary features. Similarly, Audebert et al. [26] used optical data combined with a two-stream deep network to obtain better semantic maps more rapidly. Zhang et al. [27] performed modal calibration before feature augmentation to acquire remote context semantics. The above methods produce excellent segmentation performance by fusing multimodal information using only simple addition, subtraction, and multiplication. However, they do not effectively use the potential properties of additional modalities. Faced with complex remote sensing backgrounds, unique contextual semantic extraction modules must be specifically designed to solve the problem of segmenting cars with small objects and large areas of discontinuous vegetation.

To this end, Du et al. [28] used a separated convolutional backbone to extract multimodal information and dense connectivity to improve the representation of complex scenes. Seichter et al. [29] realized an efficient and lightweight segmentation network. Gu et al. [30] proposed a hyperspectral image decomposition framework to extract and classify features. He et al. [31] designed a spatial semantic fusion module to capture shallow visual information and deep semantics under different receptive fields. Zhu et al. [32] used a self-attention mechanism to refine the cross-modal features for subsequent fusion. Yang et al. [33] explored the relationship between modal differences by studying the magnitude of the modal weights.

In contrast to previous studies, we propose the implementation of directional filter blocks (DFBs). This method performs feature enhancement at both the channel and spatial levels. First, redundant information is filtered from the channel using a channel shuffle operation. Second, a two-stage directional convolution block is used to extract multiscale contextual semantic information from the spatial level. The proposed directional convolution replaces the traditional 3×3 convolution and improves the feature extraction capability, with a reduced number of parameters and computations.

B. Prompt Injection Network

Most recent multimodal fusion networks are based on dual-stream architectures, treating IRRG features and another

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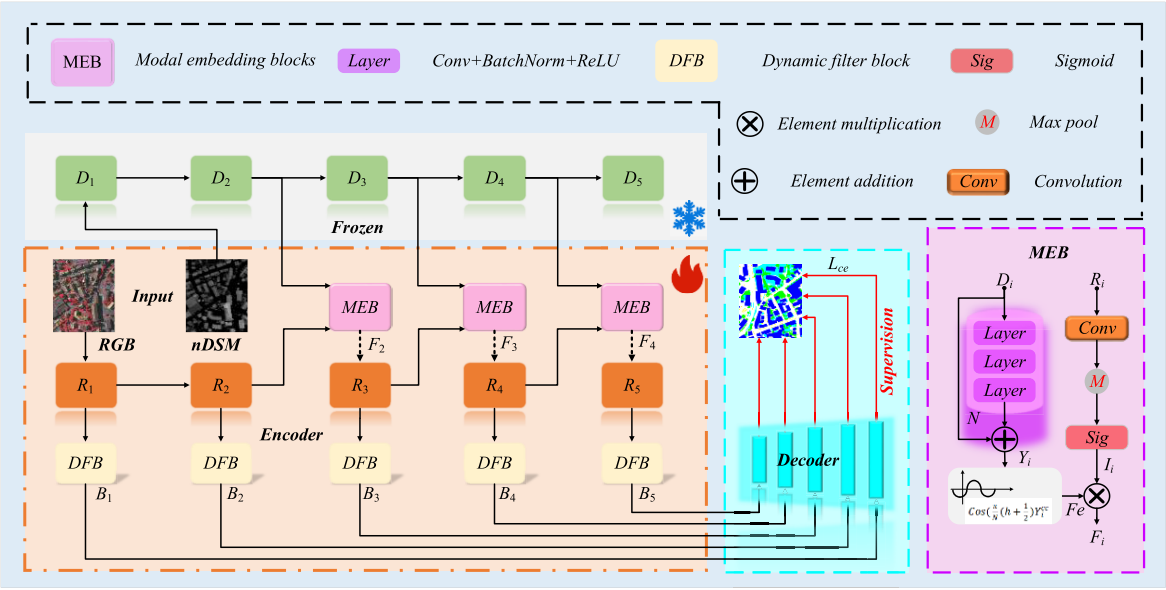


Fig. 1. Architecture of the proposed LENet. In the encoding phase, nDSM modalities are introduced using the PIM architecture, following which the MEB module is used to fuse the multimodal information. Finally, the decoding part uses DFBs for feature enhancement to obtain multiscale contextual information.

modality as independent information and extracting the features separately. Some researchers have also attempted to perform feature fusion and complementation using the same trunks to mine multimodal features simultaneously through a shared-trunk architectural approach. Inspired by audio and visual tasks, several researchers have aimed to introduce alternative modal information using prompts and complement feature information with simple addition or complex modal fusion blocks. This processing architecture has achieved significant success in the areas of classification [34], segmentation [35], and salient target detection [36], [37].

For example, Mu et al. [38] enhanced the RGB information flow by extracting important feature information as inputs from an additional high-performance network. Bu et al. [39] used pretrained models that suggest lightweight networks to reduce training time and cost. Zhuo et al. [40] used prompt learning to tune textual and visual representations efficiently and significantly reduce the computational complexity of the models. Khattak et al. [41] similarly utilized prompt learning to introduce additional modal information for enhancing visual representations. Wu et al. [42] used adaptive contextual prompts to adjust the representational space in visual sentiment analysis dynamically. Inspired by this, we present the PIM, which injects nDSM schema information. This strategy significantly reduces the training overhead and easily switches the content of the prompt information that we wish to inject to process downstream tasks more efficiently.

We embed the prompt information into the IRRG branch by freezing the backbone of the nDSM modality and extracting the middle three layers with rich semantic and detailed information from the backbone to use as prompt information. In addition, we design the MEB with few parameters for simple multimodal information fusion, as described in Section III. Using a minuscule number of parameters, this approach optimally matches complementary information between modalities to improve the segmentation performance.

III. Method

We design LENet to solve remote sensing scene-parsing tasks in complex contexts. Fig. 1 shows the overall architecture used in our approach. To minimize the number of parameters, the IRRG and nDSM features are extracted using MobileNet-V2, whereas the nDSM modal features are frozen using a prompt method to reduce the training overhead. The model architecture is relatively simple. There are three MEB sets for injecting the nDSM features of the prompt information into the IRRG branch and five DFB sets that further enhance the multiscale contextual semantics.

A. Prompt Injection Network

The proposed PIM architecture differs from the conventional two-branch encoding–decoding architecture. Prompt learning allows models to adapt to multiple tasks without additional fine-tuning, which not only significantly saves computational resources and time but also enables models to better understand and capture contextual information to generate more coherent and relevant responses. This is particularly important for complex language comprehension and generation tasks. Conventional multimodal architectures use a dual-backbone network to learn the IRRG and nDSM features separately and a fusion block to complement the feature information. To minimize the training overhead, we use prompt injection to fuse multimodal features and reinject them back into the single-backbone network for feature enhancement. Fig. 1 shows the detailed operations. Advanced nDSM features can help to localize salient objects and provide contextual semantic relationships. Therefore, we propose a prompt injection scheme that maps advanced nDSM features into the IRRG branches to help the model to generate more accurate semantic representations.

Subsequently, we design the MEB with a small number of parameters to fuse and complement the relationship between the two modalities. First, we select the middle three layers R_i and D_i ($i = 2$, 3, and 4) of features with rich semantic and

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detailed information from the IRRG and nDSM backbones as input into the modality embedding block. The details are as follows.

The MEB consists of two branches. Branch I is used to process the IRRG modal information, which is first adjusted to the channel using a convolution operation and then undergoes max pooling and sigmoid functions to filter out the essential spatial pixel weights in the IRRG modality for the subsequent multimodal fusion of the complementation

$$X_i = \mathrm{Re}(\mathrm{BN}(\mathrm{Conv}(R_i))), \quad i = 2, 3, 4 \tag{1}$$

$$I_i = \mathrm{Sig}(\mathrm{Max}(X_i)) \tag{2}$$

where $\mathrm{Sig}(\cdot)$ and $\mathrm{Max}(\cdot)$ denote the sigmoid function and max pooling operations, respectively; and $\mathrm{Conv}(\cdot)$, BN, and $\mathrm{Re}(\cdot)$ denote convolution, batch normalization, and ReLU activation, respectively.

Branch II is used to process the nDSM modal features, as shown in Fig. 1. We use several convolutional operations with parameters to capture the semantic information dynamically. We also use the DCT operation to extract the high- and low-frequency information of the nDSM modal features to capture the multispectral features effectively. Unlike the traditional channel attention approach, this DCT processes the high and low frequencies separately, and the processing operations are as follows:

$$Y_i = \mathrm{Re}(\mathrm{BN}(\mathrm{Conv}(D_i))) + D_i, \quad i = 2, 3, 4 \tag{3}$$

$$F_N = \mathrm{DCT}(Y_i) = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} \mathrm{Cos}\left(\frac{\pi}{N}\left(h + \frac{1}{2}\right) Y_i^{cc}\right) \quad (cc = C/N), \quad N \in \{0, 1, \ldots, N-1\} \tag{4}$$

where DCT denotes the DCT operation, H denotes the height of the input feature, W denotes the width of the input feature, and N denotes the frequency classification, which we initialize and set to 16.

Finally, we perform modal information complementation and feature enhancement using pixelwise multiplication operations for the two-branch features

$$\mathrm{Fe} = \mathrm{Sig}(F_N) \times I_i. \tag{5}$$

B. Directional Filter Block

Because the data acquired by different sensors may have large differences and incomplete parts even after fusion, to further enhance the branching properties of IRRG and the features after modal fusion, we also introduce DFB to characterize the features in the channel and space. Compared to the previous feature enhancement module, our dynamic filter has improved convolutional shapes for multiscale mining in both horizontal and vertical directions, and its depth-separable convolution significantly reduces the model parameter. As shown in Fig. 2, the input features are evenly distributed in the channel to obtain better and more uniform channel information. Two-stage directional filtering is performed on the features in series by the lateral and vertical dilated convolutions to obtain a wider and larger receptive field of the feature map.

Shuffler: Inspired by ShuffleNet [43] and shuffle attention [44] in the channel dimension, information exchange along the channel is achieved by shuffling features, and we use four sets of shuffles such that the direction information is evenly dispersed. We then split the features into four feature subsets along the channel dimension to generate $\{S_{i1}, S_{i2}, S_{i3}, S_{i4}\}$.

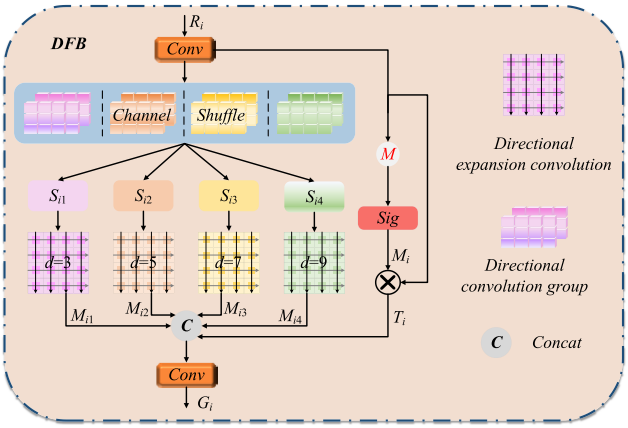


Fig. 2. Details of the proposed DFB. The channel shuffler primarily processes channel-level information, and the direction convolution unit is used to obtain multiscale contextual information.

Through the above operation, the interaction between features with different orientations is activated such that each subfeature uniformly contains orientation information in all four directions. This allows accurate spatial attention maps to be generated for each subfeature.

Directional Convolution Unit: The directional convolution unit considers two basic directions and comprises two directional convolution layers, namely, horizontal ($\mathrm{Conv}_{h,1}(\cdot)$) and vertical ($\mathrm{Conv}_{1,w}(\cdot)$), to expand the sensory field to effectively obtain the detailed information and contextual semantic relations. We use four different expansion rates on each of these two oriented convolutional layers in parallel to improve the perceptual range. This approach achieves efficient segmentation performance and reduces the number of parameters

$$M_{i1} = \mathrm{Conv}_{h,1}(\mathrm{Conv}_{1,w}(S_{i1}, d = 3), d = 3) \tag{6}$$

$$M_{i2} = \mathrm{Conv}_{h,1}(\mathrm{Conv}_{1,w}(S_{i2}, d = 5), d = 5) \tag{7}$$

$$M_{i3} = \mathrm{Conv}_{h,1}(\mathrm{Conv}_{1,w}(S_{i3}, d = 7), d = 7) \tag{8}$$

$$M_{i4} = \mathrm{Conv}_{h,1}\big(\mathrm{Conv}_{1,w}(S_{i4}, d = 9), d = 9\big) \tag{9}$$

where $\mathrm{Conv}_{h,1}(\cdot)$ denotes a horizontal convolution with a 3×1 convolution kernel size, $\mathrm{Conv}_{1,w}(\cdot)$ denotes a vertical convolution with a 1×3 convolution size, and d denotes the dilation rate of the convolution

$$M_i = \mathrm{Re}(\mathrm{BN}(\mathrm{Conv}(R_i))), \quad i = 1, 2, 3, 4 \tag{10}$$

$$T_i = \mathrm{Sig}(\mathrm{Max}(M_i)) \times R_i. \tag{11}$$

Finally, we parallelize the four branches while mining information from different directions of the features and concatenate the convolution results for information integration

$$G_i = \mathrm{Conv}(\mathrm{Cat}(M_{i1}, M_{i2}, M_{i3}, M_{i4}, T_i), \mathrm{dim} = 1). \tag{12}$$

C. Loss Function

Training the entire network is a fully supervised process. We perform pixel computation for the five output branches of

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TABLE I
QUANTITATIVE EVALUATION OF MODELS ON THE VAIHINGEN DATASET

	Imp.surf		Tree		Low veg.		Building		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S$_{2015}$[4]	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
U-Net$_{2015}$[5]	91.68	80.90	89.84	86.50	77.97	65.91	91.30	77.86	75.80	71.22	79.75	71.34
SegNet$_{2017}$[6]	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
MANet$_{2022}$[21]	91.57	83.23	93.77	90.04	76.29	65.59	91.91	77.94	86.03	81.01	87.91	79.56
ESANet$_{2021}$[29]	92.09	85.18	92.35	77.65	75.72	65.48	94.93	91.16	75.92	70.11	86.20	77.92
CMXNet$_{2022}$[27]	91.78	85.39	92.59	78.85	77.00	66.29	95.15	91.28	83.23	78.58	87.95	80.08
MGCNet$_{2022}$[33]	91.18	84.89	92.76	78.53	76.22	65.74	95.58	91.52	79.56	75.33	87.06	79.20
CMANet$_{2022}$[32]	92.02	85.06	91.48	78.54	77.35	66.06	95.29	91.15	73.56	68.89	85.94	77.94
SFAF-MA$_{2023}$[31]	90.94	85.37	90.78	78.95	80.99	67.55	95.67	92.21	74.74	71.19	86.62	79.05
EAEFNet$_{2023}$[22]	93.78	84.37	89.11	79.00	79.87	67.98	95.35	92.25	81.54	74.24	87.93	79.57
CEKD$_{2023}$[20]	92.22	85.05	92.69	78.97	76.57	66.77	95.78	91.66	76.63	72.01	86.71	78.89
Seanet$_{2023}$[23]	91.14	78.58	89.92	85.50	72.25	61.33	90.50	74.58	18.60	18.43	72.48	63.68
MAGNet$_{2024}$[16]	92.16	83.09	94.89	91.56	84.16	65.87	82.85	75.63	66.45	63.17	84.10	75.87
CAINet$_{2024}$[17]	90.50	83.10	95.16	91.22	77.28	64.94	91.16	77.76	54.11	51.07	81.64	73.62
BEDSN$_{2024}$[13]	92.30	85.71	89.24	80.56	83.71	71.99	94.71	89.95	86.31	75.92	89.25	80.82
LENet	93.17	85.32	90.79	79.57	80.06	68.53	95.82	93.32	87.93	83.73	89.56	82.10

the decoding part with the ground truth (GT) using only the simple cross-entropy loss

$$L_{\text{sup}} = \sum_{i=1}^{5} L_{\text{ce}}(G_i, \text{GT}) \tag{13}$$

$$L_{\text{ce}} = -\sum_{i}^{K} Y_i \times \log(P_i), \quad i = 0, 1, 2, 3, 4 \tag{14}$$

where $L_{\text{ce}}(\cdot)$ represents the cross-entropy loss, Y_i is a binary indicator variable, L_{sup} signifies the supervised utilization of the GT to monitor the output of each decoding block, K denotes the total number of classes, and P_i represents the predicted probability of an observed sample belonging to class i.

IV. EXPERIMENTAL RESULTS

A. Datasets and Evaluation Metrics

1) Datasets: The publicly available Vaihingen and Potsdam remote sensing image segmentation datasets were used to validate the proposed network. The datasets have six main categories that primarily segment the five categories of low vegetation, buildings, cars, pavements, and trees that are common to every city. Another category of backgrounds (e.g., containers, swimming pools, and tennis courts) includes semantic objects that are usually considered of no interest in an urban scene. In addition, we explored RGB-D mirror segmentation [45] of the indoor dataset to validate the robustness of the model further.

2) Metrics: We used four metrics to assess the value of each classification result: the intersection over union (IoU), class accuracy (Acc), mean IoU (mIoU), and mean Acc (mAcc). The mirror segmentation dataset was evaluated for segmentation accuracy using four metrics: the IoU, F-measure (F_β^W), mean absolute error (MAE), and balanced error rate (BER).

B. Experimental Setup

PyTorch was used as the framework to implement the model throughout the training process. We also utilized an NVIDIA Titan processor for accelerated computation. Adam was used for optimization and the momentum was configured to 0.9. The initial learning rate was set to 0.0001 and the momentum parameter was decayed by 0.1× every 20 epochs during training.

C. Performance Evaluation

To evaluate the effectiveness and robustness of LENet further, we compared it with current, more advanced methods. Notably, LENet outperformed the comparison methods in most metrics on both datasets, as shown in Tables I and II. To validate the effectiveness of our modules, Table III further demonstrates the complexity analysis of the proposed model, showing that our model is lightweight and efficient. Table IV also further validates the robustness of the proposed network. It can be observed from the results that its segmentation performance was not very different from that of the model with high complexity.

1) Quantitative Evaluation: As indicated in Tables I and II, LENet achieved good segmentation results on both datasets while maintaining lightness and efficiency. In the tables, red indicates the optimal value and green indicates the second optimal value. The proposed PIM reduced the overhead by half while effectively injecting multimodal information into the IRRG branches during training. According to the ablation results presented in Table III, the proposed DFB clearly captured semantic contextual relationships at both large and small scales, effectively segmenting cars, vegetation, and buildings set against complex backgrounds. Finally, the data in Table III further show that when comparing LENet with current state-of-the-art lightweight networks, the former completely outperformed other models in terms of the number of parameters, performance, and FPS.

2) Qualitative Evaluation: We selected five groups of images for comparison. The overall comparison results are shown in Fig. 3. Notably, our lightweight model still achieved a relatively good segmentation effect under low parameter conditions. Simultaneously, its segmentation effect was considerably better than that of the lightweight model under the same backbone conditions. Our lightweight model still selected useful complementary information for supplementation using the PIM and overcame the interference caused by complex backgrounds. The ablation result plots show that

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TABLE II
QUANTITATIVE EVALUATION OF MODELS ON THE POTSDAM DATASET

	Imp.surf		Tree		Low veg.		Building		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S$_{2015}$[4]	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
U-Net$_{2015}$[5]	90.03	80.27	88.71	82.92	85.82	71.60	84.06	72.05	93.89	90.24	50.30	36.26	82.13	72.22
SegNet$_{2017}$[6]	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
MANet$_{2022}$[21]	90.48	81.20	87.58	83.47	88.78	72.74	82.69	71.54	90.96	88.73	46.16	38.11	81.11	72.63
ESANet$_{2021}$[29]	91.38	82.92	82.48	70.81	87.10	73.16	93.69	89.82	93.08	88.53	55.68	43.38	83.90	74.77
CMXNet$_{2022}$[27]	90.54	81.86	83.52	71.45	87.17	73.11	93.80	89.28	92.23	87.99	51.61	40.58	83.14	74.04
MGCNet$_{2022}$[33]	90.61	81.21	83.70	72.51	86.72	72.67	93.87	89.02	93.39	88.91	53.24	39.66	83.59	74.00
CMANet$_{2022}$[32]	91.75	81.87	84.70	72.40	86.42	73.59	93.64	90.23	92.74	89.11	55.97	42.74	84.20	74.99
SFAF-MA$_{2023}$[31]	90.63	82.06	83.51	70.95	85.76	72.35	93.31	89.67	92.55	88.34	60.03	42.40	84.30	74.30
EAEFNet$_{2023}$[22]	92.32	82.04	85.02	71.91	84.34	72.37	93.22	89.54	94.64	88.30	53.55	39.34	83.85	73.92
CEKD$_{2023}$[20]	91.09	80.89	85.18	71.62	84.70	71.71	93.37	88.97	93.01	89.35	52.63	38.18	83.33	73.45
Seanet$_{2023}$[23]	89.80	78.77	88.34	81.99	86.87	69.48	74.97	63.17	90.37	86.52	42.13	34.30	78.75	69.04
MAGNet$_{2024}$[16]	89.51	80.26	92.76	89.13	83.20	70.51	83.19	65.17	94.08	83.16	51.76	33.98	82.42	70.37
CAINet$_{2024}$[17]	90.52	80.35	93.13	89.32	84.70	70.55	81.15	67.89	93.48	85.33	48.06	37.04	81.84	71.75
BEDSN$_{2024}$[13]	91.83	84.90	86.72	76.55	85.85	75.20	95.64	91.64	95.03	90.53	49.42	32.82	84.08	75.27
LENet	92.42	83.06	83.78	73.22	87.35	74.35	93.70	90.00	94.91	88.34	55.53	42.67	84.62	75.27

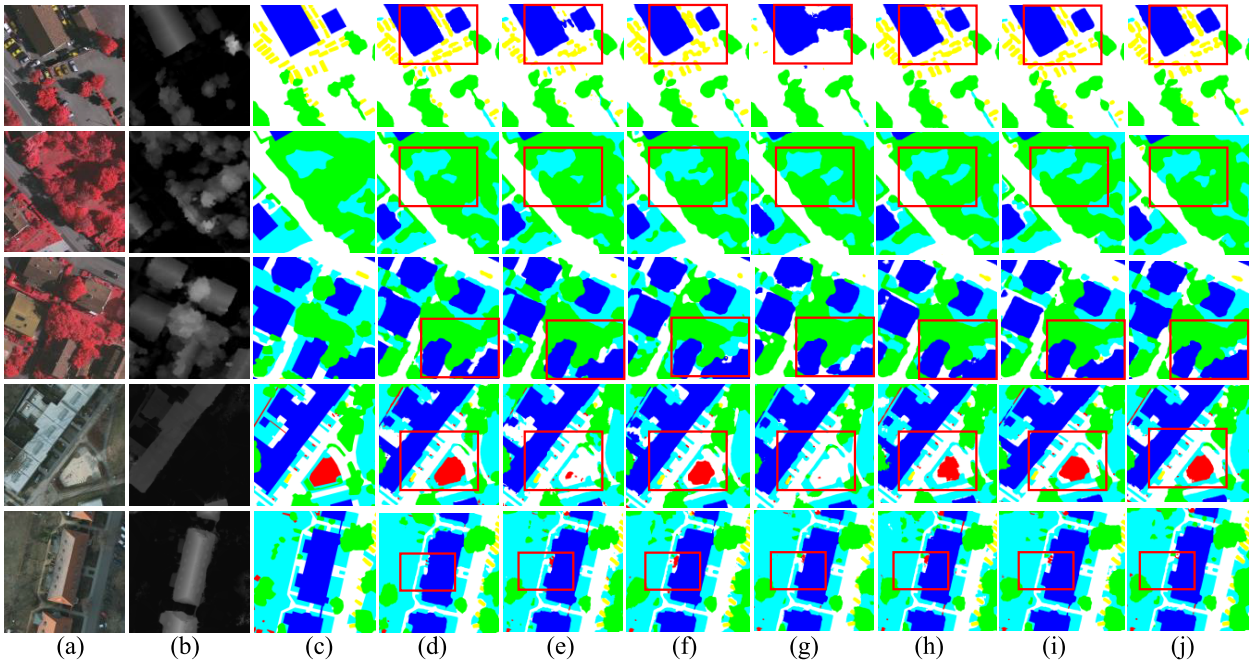


Fig. 3. Comparative segmentation results. (a) IRRG, (b) nDSM, (c) GT, (d) ours, (e) CAINet, (f) MAGNet, (g) SeaNet, (h) CEKD, (i) EAEFNet, and (j) CMANet.

TABLE III
ABLATION EXPERIMENTS FOR EACH MODULE OF OUR NETWORK

Methods	PIM	MEB	DFB	VAIHINGEN		POTSDAM	
				Macc	Miou	Macc	Miou
Wo/PIM			√	80.37	72.68	83.58	74.27
Wo/MEB		√	√	86.41	79.03	84.47	74.61
Wo/DFB	√		√	85.55	77.98	84.09	74.44
Ours	√	√	√	88.91	81.47	84.77	75.69

the proposed DFB reduced the module parameters and added multiple sensory field range features. This improved the results for categories involving small receptive field captures, such as small cars, large continuous irregular areas of vegetation, and buildings.

3) Complexity Comparison: We propose a lightweight model for remote sensing scene applications for effective deployment on mobile devices. However, pursuing a lightweight network alone is insufficient because good segmentation performance should also be considered. Table V lists the number of parameters, computation, and FPS of LENet and compares these results with existing lightweight and high-performance networks. The segmentation performance of LENet significantly exceeded that of a 200 M-sized network, with only 10.87 M parameters. Its segmentation performance was considerably better than that of the same type of lightweight networks.

In real-world remote sensing applications, small vehicles are popular owing to their portability, ease of deployment, flexibility, and low cost. However, these vehicles often have a limited operational range and computational power. Our lightweight model, LENet, maintains high-quality segmentation results with a low number of parameters and minimal computational requirements, making it suitable for rapid deployment on small mobile devices with limited computing and storage capacity.

D. Ablation Study

1) Effectiveness of PIM: Table III and Fig. 4 illustrate the effectiveness of our PIM. Our PIM reduced the training

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TABLE IV
QUANTITATIVE EVALUATION OF MODELS ON THE RGB-D MIRROR DATASET

	Methods	IoU↑	BER↓	MAE↓	F_β^ω↑	Params/M↓	FLOPs/G↓
RGBD SOD RGBD Minor segmentation	BBSNet $_{2020}$ [45]	62.16	14.39	0.089	0.648	49.78	43.85
	SPNet $_{2021}$[46]	66.77	11.64	0.068	0.697	118.92	64.8
	CIRNet $_{2022}$[47]	62.87	15.53	0.068	0.674	103.15	59.40
	CAVER $_{2023}$ [48]	63.37	13.74	0.074	0.696	93.78	57.66
	CATNet $_{2024}$ [49]	63.82	14.18	0.063	0.696	272.1	171.2
	CPNet $_{2024}$ [50]	62.64	13.48	0.077	0.714	223.1	125.35
RGBD -Semantic Segmentation	RFNet $_{2020}$ [51]	59.87	10.30	0.606	0.103	23.67	14.86
	ESANet $_{2021}$ [52]	68.29	11.81	0.066	0.720	54.46	34.82
	CAINet $_{2024}$[16]	74.93	10.42	0.047	0.428	13.8	21.83
	SANet $_{2022}$[53]	78.43	8.16	0.041	0.834	104.80	66.10
	SATNet $_{2023}$ [54]	78.42	9.31	0.031	0.836	108.29	156.74
	LENet	75.26	9.20	0.048	0.828	10.81	2.72

TABLE V
COMPUTATIONAL COMPLEXITY COMPARISON

Model	FPS	Params (M)↓	FlOPs ↓
FCN-8S$_{2015}$[4]	51.5	26.36	55.93
U-Net$_{2015}$[5]	82.4	134.29	74.55
SegNet$_{2017}$[6]	37.45	53.56	18.3
MANet$_{2022}$[21]	37.45	35.86	19.49
ESANet$_{2021}$[29]	17.91	45.42	10.15
CMXNet$_{2022}$[27]	31.69	66.56	14.26
MGCNet$_{2022}$[33]	10.84	209.99	47.51
CMANet$_{2022}$[32]	12.12	117.82	28.61
SFAF-MA$_{2023}$[31]	58.85	232.08	18.24
EAEFNet$_{2023}$[22]	34.33	77.97	23.3
CEKD$_{2023}$[20]	18.24	245.03	68.61
Seanet$_{2023}$[23]	103	2.76	1.33
MAGNet$_{2024}$[16]	41.2	16.58	4.39
CAINet$_{2024}$[17]	68.66	13.8	4.67
LENet	61.09	10.81	2.72

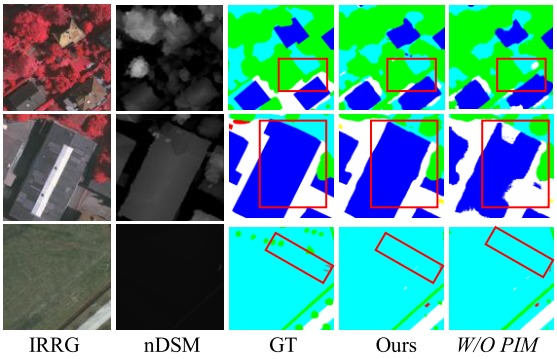


Fig. 4. Comparison of the visual results of our method and W/O PIM.

overhead while maintaining good segmentation performance compared with the unimodal scene parsing of remote sensing images and the two-branch multimodal approach. The ablation results presented in Fig. 4 further demonstrate that our prompt injection method effectively supplemented the modal information in the RGB branch compared with the noninjection architecture. As evidenced by the first row in Fig. 4, the modal injection method could distinguish the relationship between vegetation and buildings. This indicates that, by using a small number of parameters, our proposed PIM successfully embeds the superior features of another modality into the RGB branch, effectively overcoming the problem of poor segmentation performance arising from complex backgrounds.

2) Effectiveness of the Proposed DFBs: As shown in Table III and Fig. 5, DFBs form part of our feature enhancement strategy. We evaluated the proposed LENet by

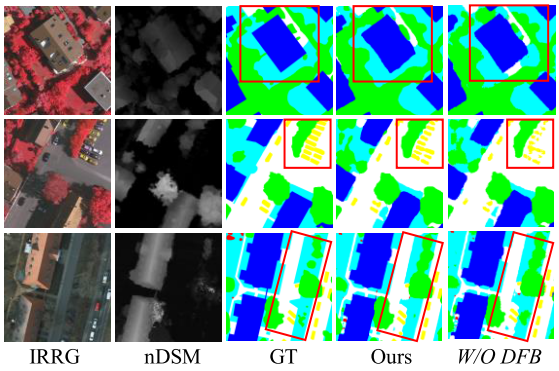


Fig. 5. Comparison of the visual results of our method and W/O DFB.

eliminating the DFBs to demonstrate their effectiveness. First, we eliminated the DFBs and directly fed the output features from the backbone network into the decoding block. The prediction results of the model differed significantly from the GT diagram in complex scenarios; for example, the relationship between buildings and vegetation was intertwined, the segmentation of small object categories such as cars was often inaccurate, and the edges of buildings were not well defined. The addition of DFBs may prevent this phenomenon. These examples prove the effectiveness of the proposed DFBs in extracting local and global contextual semantic information and better dealing with complex backgrounds.

3) Effectiveness of the Proposed MEB: To fuse and complement the relationship between the bimodal source features more effectively, we processed them separately to reduce the differences. The ablation results presented in Table III and Fig. 6 indicate that when the MEB of LENet was eliminated, the model prediction results exhibited notable deficiencies, such as missing information. For example, in Fig. 6, there is a significant disparity in information between low vegetation and buildings in the first row, incomplete predictions of building information in the second row, and information missing from the background classes in the third row. This experiment clearly demonstrates the accuracy of our proposed MEB module, which effectively incorporates additional modal information into the new RGB branch.

E. Performance on Other Datasets

To demonstrate the robustness and generalization ability of our proposed LENet model further, we conducted in-depth experimental validation on the RGB-D mirror dataset. The dataset consists of 3049 pairs of RGB images and their

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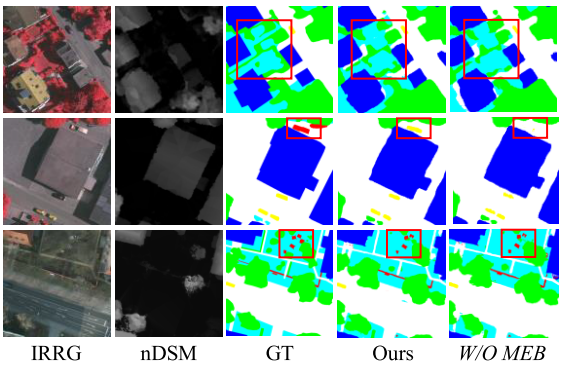


Fig. 6. Comparison of the visual results of our method and W/O MEB.

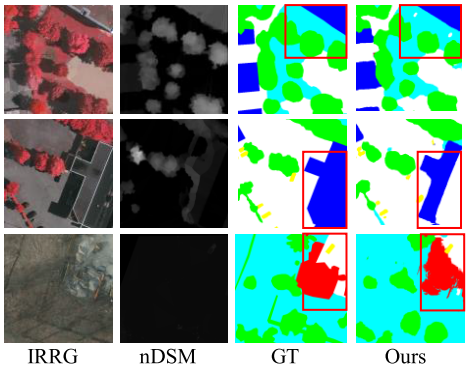


Fig. 7. Examples of failures of the proposed LENet.

corresponding depth images, which are specifically divided into 2000 training samples and 1049 test samples, providing a comprehensive database for evaluation. During the performance evaluation, we used four key metrics to measure the segmentation accuracy: the IoU, F_β^W, MAE, and BER. It can be clearly observed from the results in Table IV that the proposed LENet model outperformed the other compared methods in all four metrics. This not only reflects the strong adaptability of LENet in dealing with complex visual tasks but also further confirms its excellent robustness and good generalization properties.

F. Failure Examples

Although the proposed LENet achieved good segmentation performance, it struggled to predict certain challenging scenarios accurately owing to parameter constraints. As shown in Fig. 7, the intricate segmentation of scenes involving alternating vegetation and buildings exhibited significant segmentation deviations. In addition, the accuracy of category segmentation for the background was relatively low. This is primarily attributed to the complex background composition, which included uncommon categories found in some cities, making it difficult for the model to learn and segment background category information effectively. Consequently, this resulted in a large area of confusion. This presents a challenge for existing algorithms, and our only recourse is to minimize these deviations. These observations indicate that our LENet still has several limitations and the segmentation accuracy for certain categories remains inadequate in extremely complex scenes.

V. Conclusion

We investigated LENet as a method for parsing complex remote sensing background images. In the evaluation, the training overhead of our PIM differed insignificantly from that of unimodal networks, and its segmentation results were considerably superior to those of state-of-the-art unimodal methods. Furthermore, it achieved comparable results to those of a larger model, despite having fewer parameters than the bimodal network. The experimental plots and ablation results demonstrated that the PIM effectively fuses and complements information from another modality into the IRRG branch, thereby successfully managing complex backgrounds. Considering the inconsistent size and irregularity of the category information in segments of remote sensing images, we designed directional filters to extract features such as cars, vegetation, and buildings. Multiple parallel multidirectional sensory field convolutions were introduced to extract multiscale features and explore contextual semantic relationships between the local and global features. Finally, we verified the efficacy of the proposed framework and the feasibility of the proposed directional filters through extensive experiments. In the future, we aim to improve the generalization of the PIM to serve as a lightweight and effective tool for multiple scene-parsing applications. Furthermore, we will attempt to apply LENet to other scene-parsing tasks to validate the robustness of the model further. We also plan to deploy the model on small mobile devices to provide realistic feedback regarding its performance in real-world applications.

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